

Research Paper

Theoretical calculation and experimental analysis on humidification performance of high-performance water-absorbing porous fiber plates

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ABSTRACT

In order to improve the indoor relative humidity in winter, a high-performance water-absorbing porous fiber plate was introduced in this paper, which absorbed water and infiltrated through capillary action, and an experimental system with porous plates arranged in parallel as humidifying elements was designed and constructed to explore the humidifying performance of porous plates and to expand its application field. The experimental results showed that the humidification effect of porous plates was excellent. At a relative humidity of 15 %, mainstream air velocity of 3.5 m/s, and air temperature of 45 °C, the maximum humidification capacity of the system reached 266.4 g/h. The humidification capacity was proportional to the air temperature and mainstream air velocity, and inversely proportional to the relative humidity of the air. The mainstream air velocity increased by 0.5 m/s, the humidification capacity increased by about 20 g/h. The change trend of humidification capacity and heat transfer quantity calculated through the criterion correlation was consistent with the measured values, and the relative errors between the theoretical and experimental values for most of the conditions were within 20 %. One of the reasons for the errors might be that the mainstream air velocity in the gaps between the porous plates was greater than the mainstream air velocity of the experimental air duct used in the theoretical calculation. Due to the difficulty in measuring the air velocity in the gaps between porous plates, CFD simulations were conducted. The relative errors between the theoretical corrected values calculated with the simulated air velocities and the experimental values were within 10 % for most of the conditions, and between 10 % and 15 % for very few conditions.

1. Introduction

In modern buildings, people often pursue a healthy and comfortable indoor thermal environment. Ambient humidity plays a crucial role in indoor air quality [1], human thermal comfort [2] and health [3]. Lower indoor air humidity enhances the survivability, transmission and contagiousness of influenza viruses [4,5], leads to respiratory illnesses [6,7], affects eye health [8,9], increases work stress [10], reduces work efficiency [11], and generates static electricity [12]. Considering the importance of humidity, environmental humidity management has been introduced to ensure a healthy and safe living environment and to fulfill the needs of certain production processes [13].

Research has shown that various humidification methods can effectively control environmental humidity. For example, ultrasonic humidification [14], dry steam humidification [15], centrifugal humidification [16], rotary humidification [17], wet membrane

humidification [18], etc. However, ultrasonic humidification simultaneously generates water vapor and small droplets composed of water and impurities. Due to the action of gravity, most of the water droplets move downwards and gradually evaporate, resulting in stratification in the temperature and humidity field [19]. Moreover, ultrasonic humidifiers have certain requirements for water quality [14,20]. Dry steam humidification must have a stable steam source and be accompanied by gas pipelines [21]. Centrifugal humidification may produce large water droplet particles that cannot be completely evaporated, resulting in lower humidification efficiency [21]. The rotary humidification is not suitable in extremely cold areas with extremely low relative humidity [22]. Evaporative humidifiers with porous membranes as the core component have been widely used due to their advantages of low energy consumption, simple process, and wide humidification range [13,18]. Dry air passes through the soaking element (porous membrane) of the humidifier, and water absorbs heat and evaporates as the mainstream air enters the environment, increasing the humidity in the air [23,24].

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Nomenclature			
A	Area of the porous plates (m^2)	Pr	Prandtl Number
C_a	Concentration of water vapor in the air (mol/m^3)	Q_c	Convective heat transfer quantity (W)
C_p	Concentration of water vapor on the porous plates (mol/m^3)	Q_e	Heat quantity absorption by evaporation of water (W)
D	Diffusion coefficient of water molecules in air (m^2/s)	r	Latent heat of vaporization of water (kJ/kg)
D_0	Diffusion coefficient of water molecules in air at $T_0 = 273 \text{ K}$ (m^2/s)	Re	Reynolds number
h	Average convection heat transfer coefficient ($\text{W}/(\text{m}^2 \cdot \text{K})$)	R_w	Gas constant for water vapor ($\text{J}/\text{kg} \cdot \text{K}$)
h_m	Average convective mass transfer coefficients (m/s)	Sc	Schmidt number
H	Humidification capacity calculated by weighting method (g/h)	Sh	Sherwood number
l	Length of porous plate (m)	T_a	Air temperature before the humidifying element (K)
M	Molar mass of water vapor (kg/mol)	T_e	Air temperature after the humidifying element (K)
m_w	Mass diffusion flux (kg/s)	T_m	Average temperature of air in the concentration boundary layer (K)
Nu	Nusselt number	T_p	Water temperature on the porous plate (K)
P_a	Water vapor partial pressure at temperature T_a and relative humidity φ_a (Pa)	u	Mainstream air velocity (m/s)
P_{ps}	Saturated water vapor partial pressure at temperature T_p (Pa)	W_1	Initial indication of the weighing module (g)
		W_2	End indication of the weighing module (g)
		ν	Air viscosity (m^2/s)
		λ	Thermal conductivity ($\text{W}/(\text{m} \cdot \text{K})$)
		Δt_m	Logarithmic mean heat transfer temperature difference (K)

An ideal porous membrane needs to quickly absorb water and then diffuse it with a wider moist area to achieve rapid evaporation. Similar to fabrics or textiles, micropores have a good ability to diffuse water through capillary action [25]. Chen [25] et al. compared five evaporation materials through experiments on wicking height, diffusivity, and moisture absorption and desorption, and the results showed that the plant fiber polymer composite had better moisture absorption performance than other selected materials. Xu [26] et al. tested a series of fabrics woven from various fibers, and the experimental results showed that compared to kraft paper, most fabrics have excellent performance in terms of moisture absorption, diffusion, and evaporation. Kim [27] et al. used PET/cellulose coated sheet (50/50 % by mass) as a wet channel material, and the results showed that it had excellent wetting properties. Su [28] et al. used profiled polyester fibers and natural cotton fibers in different blending ratios to spin composite yarns, and tested their water absorption, diffusion, and drying properties. The experimental results showed that the use of profiled polyester fibers can improve diffusion and drying rates, achieving good water release ability.

The combination of humidifying elements with porous membranes as the core and air conditioning units is a widely used system form. The materials used for humidification elements mainly include natural fiber-based materials, stone-based materials and organic synthetic materials. Natural fibers are mainly derived from natural resources [29] and have high porosity, permeability and water retention [30]. Doğramacı [30] et al. conducted experimental studies on five natural porous materials in wind tunnels, and the results showed that eucalyptus fiber had the best effect, with relative humidity increasing from 20 % to over 60 % at an air velocity of 0.1 m/s. Stone-based materials are durable under atmospheric conditions and resistant to chemical degradation, especially when they are exposed to outdoor environmental conditions [30]. Due to the porous structure of some stones, such as siliceous shale [31], porous ceramics [32], volcanic ash [33], porous clay [34], and pumice [35], they can hold large amounts of water. Bada [34] et al. implemented a passive cooling system consisting of air pipes made of porous materials (fired clay) buried in moist soil, and the results showed a 35 % increase in relative humidity when the pipe was only 1.5 m long. In recent years, organic synthetic materials have been extensively studied. Lv [36] et al. designed and constructed a novel unit dew-point evaporative cooler with fiber membrane, and studied the effects of three air state parameters (temperature, relative humidity, and air velocity) on wet bulb efficiency. Chen [13] et al. reported biomimetic micro nano fibrous composite membranes with a highly aligned fibrous structure

using a humidity induced electrospinning technique for high-efficiency indoor humidification, demonstrating good moisture absorption height and excellent water absorption. The experimental results showed that the humidification ability of the membrane was 57 % higher than that of the non-woven commercial humidification membrane. Bishoyi [37] et al. analyzed the humidification performance of Honeycomb and Aspen cooling pads with the same rectangular cross-sectional area. The experimental results showed that both cooling pads could increase relative humidity to over 55 %, and Honeycomb cooling pads could achieve comfortable humidity in a shorter time. Nada [38] et al. conducted experimental research using the corrugated cellulose papers with bee-hive structure as humidifying elements, and the results showed that increasing air temperature, water temperature, and water flow rate resulted in improved humidifying efficiency and evaporation water volume.

It can be found that the use of appropriate humidifying materials can effectively enhance the humidity of the environment. All the humidifying materials have a pore structure with good water absorption potential. However, the humidification elements were mostly installed in the air handling unit for centralized air treatment. The water used for humidification was mostly sprayed from the top onto the humidification elements. Considering factors such as safety, more components, and larger unit size, there were few humidifying elements combined with fan coils for decentralized air treatment. This paper introduces a high-performance water-absorbent porous fiber plate, which only requires a water pan below without a top spray, and the micropores on the fiber plate surface promote capillary absorption, which then diffuses to the fiber plate surface for evaporation. The fiber plate has no moving parts and is suitable for miniaturized production. It can be placed into the fan coils rather than air handling units to provide humidification capacity while meeting the original heating capacity. This article aims to expand the application field of porous fiber plate and clarify the influencing factors of porous plate humidification. The humidification elements arranged in parallel with porous plates were designed, and an experimental system was constructed. The second part of this article introduced the experimental system, experimental materials, and experimental process. The third part mainly analyzed the influence of different inlet air parameters (temperature, relative humidity, and air velocity) on the humidification capacity of the system and the difference between theoretical and experimental values.

2. Experiments

2.1. Experimental system

In order to investigate the humidification performance of the porous plate, an experimental system as shown in Fig. 1 was constructed. The experimental system mainly consisted of a constant temperature and humidity control box, a static pressure box, a test section and a piping system. There was an electric heater (2) inside the constant temperature and humidity control box (1) for heating, connected to a refrigerant dryer (3) for cooling and dehumidification, and connected to a humidifier (4) for humidification to create a constant temperature and humidity environment. The static pressure box (5) was equipped with a supplemental heater (6) to stabilize the temperature of the air entering the test section. The test section mainly included insulation air duct (7), experimental air duct (8), uniform flow plates (9), baffle plates (10), porous plates (11, experimental materials), water pan (12, with heating pads inside), weighing module (13, under the water pan), temperature and humidity sensors and air velocity transmitters. The insulation air duct was mainly used for insulation of experimental air ducts that could not be wrapped with insulation materials. The uniform flow plate allowed the airflow to blow smoothly through the porous plates. The baffle plates were mainly used to block the airflow and allowed all the air to enter the experimental air duct. The ducting system included air supply ducts (15), return air ducts (16) and bypass ducts (17), and all the ducts were well insulated. The physical drawing is shown in Fig. 2.

2.2. Experimental materials

The porous plate was a water absorbing material made of polyester fibers, supplied by UNITIKA. The image obtained under a scanning electron microscope is shown in Fig. 3. From Fig. 3, it can be seen that the fiber diameter was uniform and there were numerous pores between the fibers, which enhanced the water absorption capacity of the porous plate.

Capillary absorption refers to the movement of water through capillary tubes in a material caused by surface interactions between the water and the pore walls in the absence of external hydraulic head

application [39]. In order to evaluate the wicking height of the porous plate used in this article, the temperature was controlled at 21.6 °C and the relative humidity was controlled at 50 % according to the experimental conditions in reference [40]. The porous plate was cut into 350 mm × 35 mm strips and hung vertically, and the bottom 25 mm was placed into a water-filled container. The comparison of the wicking height between the material used in this article and other materials tested by Chu [40] et al. is shown in Fig. 4. From Fig. 4, it can be seen that the wicking height of the fiber material used in this article was significantly higher than other materials, reaching 72 mm within 60 s. During the humidification process, water continuously evaporates, requiring the porous plate to continuously absorb water. A significant water absorption rate could ensure that the porous plate was always in a saturated state.

The humidification element composed of 20 porous plates is shown in Fig. 5. The plate thickness was 2 mm, and the plate pitch was 8 mm. A 100 mm wide rectangular hole was reserved under the experimental air duct, and the porous plates were placed in parallel arrangement in the experimental air duct, and the lower part of the porous plates were immersed in the water pan with a wetting depth of 30 mm to ensure stable water absorption of the porous plates. The organic glass plate was used to fix the porous plates and avoided the influence of water evaporation in the water pan on the experimental results. Three thermocouples were uniformly arranged inside the porous plate to monitor the water temperature. During the experiment, always kept the bottom of the porous plates soaked in water.

2.3. Experimental operation process

The key to the experiment lies in creating a constant temperature and humidity environment. Before the experiment begins, close the air valve of the supply duct, open the air valve of the bypass duct, turn on the electric heater, refrigerant dryer, and humidifier, and obtain a stable initial environment of constant temperature and humidity through PID control. At the same time, sufficient experimental water is pre-set in the water pan to ensure that the bottom of the porous plates is always in the water during the experiment. There are two heating pads inside the water pan, which maintains a constant water temperature through a PID

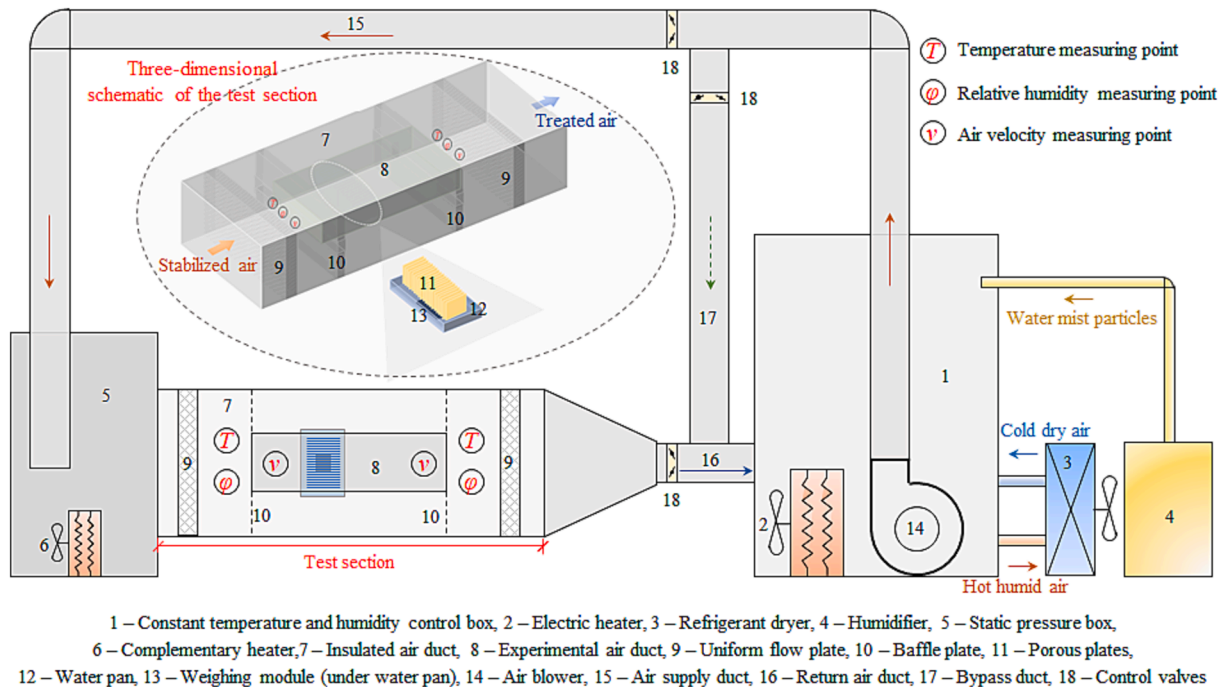


Fig. 1. Schematic diagram of the experimental system.

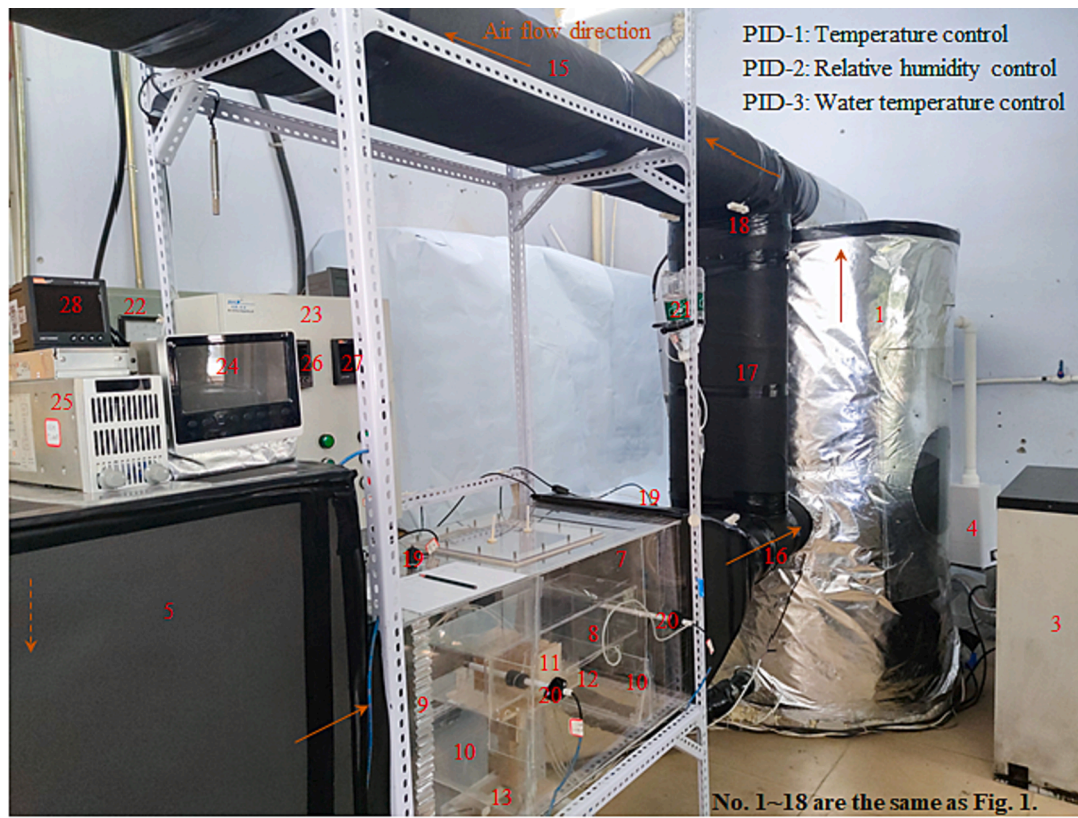


Fig. 2. Physical diagram of the experimental system.

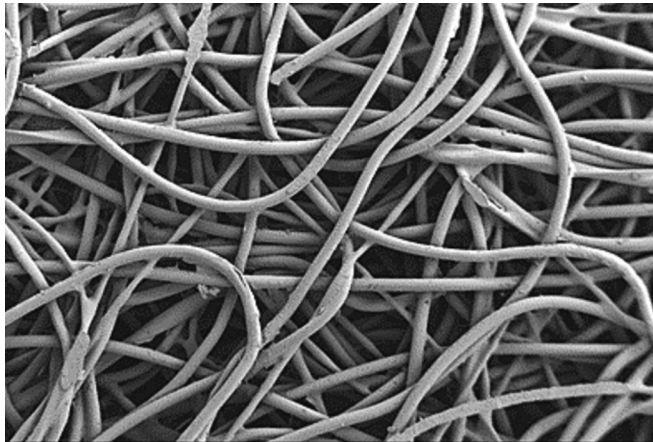


Fig. 3. Scanning electron microscope image of the porous plate (Magnification 60 times).

controller. After the preparation work is completed, close the bypass duct air valve, open the supply duct air valve, and adjust the frequency converter to change the fan speed to achieve the required air velocity value. During the experiment, the electric heater, refrigerant dryer, and humidifier were continuously in operation through PID regulation to ensure stable temperature and humidity of the air flow through the humidifier element. The test values of the working conditions at 40 °C, 20 %, and 2.0 m/s are shown in Fig. 6. From Fig. 6, it can be seen that under the condition of one-hour continuous operation, the temperature variation range was ± 0.5 °C, the relative humidity variation range was ± 2 %, and the air velocity variation range was ± 0.15 m/s. The

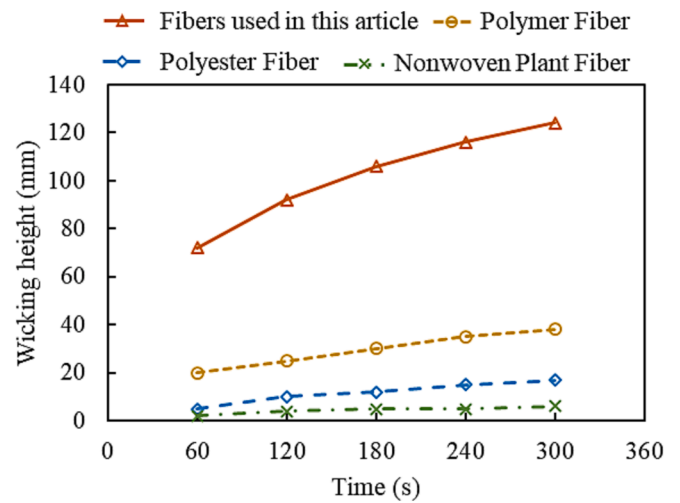


Fig. 4. Comparison of wicking height of different materials.

experimental system can create stable temperature, humidity, and air velocity conditions.

2.4. Data acquisition

In order to obtain experimental data, some experimental instruments were used during the experiment. As shown in Figs. 1 and 2, thermal resistors and humidity transmitters were installed before and after the humidification element to monitor the temperature and humidity of the air, and air velocity transmitters were installed to monitor the

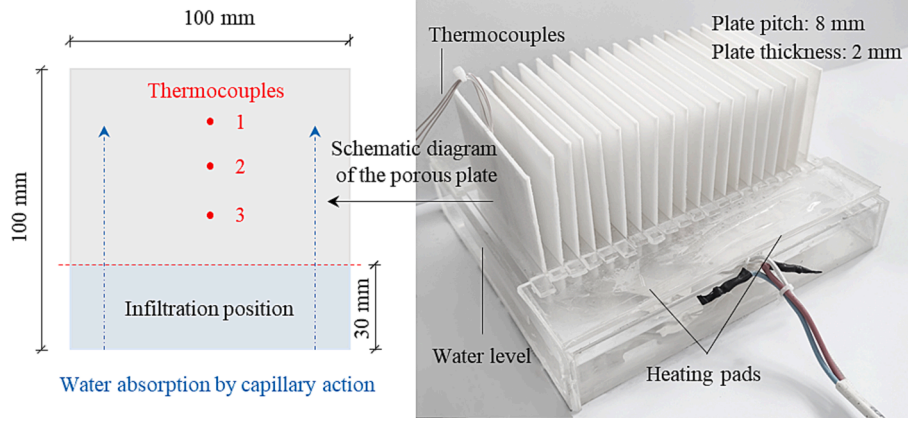


Fig. 5. Humidifying element.

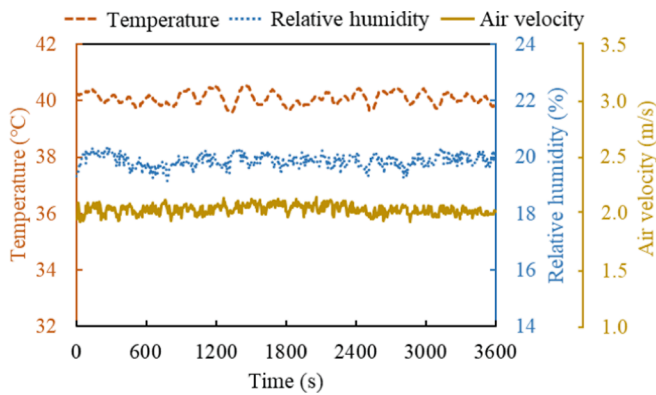


Fig. 6. Air parameters before humidifying element (under working conditions of 40 °C, 20 %, 2.0 m/s).

mainstream air velocity of the experimental air duct. There was also a thermal resistor inside the water pan to measure water temperature, and thermocouples were arranged at the upper, middle, and lower positions of the porous plate (as shown in Fig. 5) to monitor the water temperature on the porous plate. The experimental thermal resistance and thermocouple were calibrated in an oil bath before installation. A weighing module was installed below the water pan to measure the humidification capacity. All measuring instruments mentioned above can output DC or voltage signals and transmit them to the data acquisition unit for recording. The data collection device records all measurement data during the entire process at a time interval of 6 s.

The humidification capacity H is calculated according to Eq. (1) using the weighing method.

$$H = W_1 - W_2 \quad (1)$$

where, W_1 and W_2 are the initial and end indication of the weighing module.

The accuracy of each measurement instrument is detailed in Table 1.

Table 1
The accuracy of measurement instrument.

Measurement instrument	Measurement range	Measurement accuracy
T-type thermocouple	−50 – 140 °C	±0.1 °C
PT 100 thermal resistance	−50 – 140 °C	±0.1 °C
Humidity transmitter	0 – 100 %	±1.5 %
Air velocity transmitter	0 – 5 m/s	±0.05 m/s
Weighing Module	0 – 3000 g	±0.1 g

3. Results and discussion

3.1. Analysis of measured humidification capacity

The variation of the net weight of water in the water pan with time is shown in Fig. 7. It can be seen from Fig. 7 that in a particular working condition, the net weight of water with time showed an obvious linear decreasing relationship ($R^2 = 0.9999$), and the evaporation process of water was continuous and stable, which further explained that the air temperature, relative humidity, and air velocity in front of the humidification element were stable (Fig. 6). The trend of the remaining conditions was consistent with Fig. 7 and would not be repeated here.

The variation trend of humidification capacity under different mainstream air velocity and air temperature at the same relative humidity (15 %) is shown in Fig. 8. From Fig. 8, it can be seen that under certain relative humidity and mainstream air velocity, the humidification capacity increased with the increase of air temperature, but the change was not significant with temperature. For every 5 °C increased in temperature, the humidification capacity increased by about 10 g/h. When the relative humidity and temperature were constant, the humidification capacity increased with the increase of mainstream air velocity. The mainstream air velocity increased by 0.5 m/s, and the humidification capacity increased by about 20 g/h. The humidification capacity at a mainstream air velocity of 3.5 m/s was approximately 1.7 to 1.9 times that at an air velocity of 1.0 m/s. At a relative humidity of 15 %, mainstream air velocity of 3.5 m/s, and air temperature of 45 °C, the maximum humidification capacity of the system reached 266.4 g/h.

The variation trend of humidification capacity under the same temperature (40 °C), different mainstream air velocity, different relative

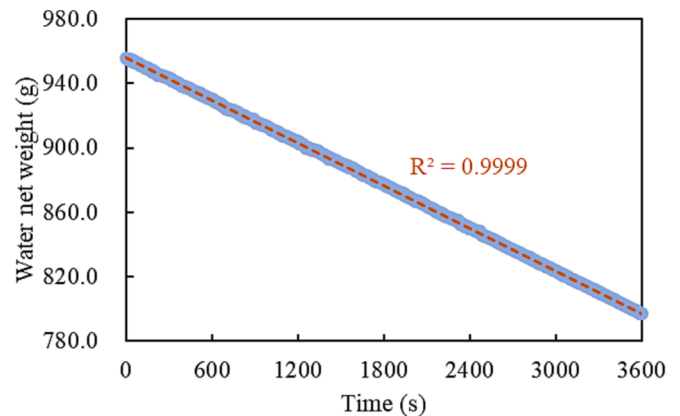


Fig. 7. The net weight of water with time (under the working conditions of 40 °C, 20 %, 2.0 m/s).

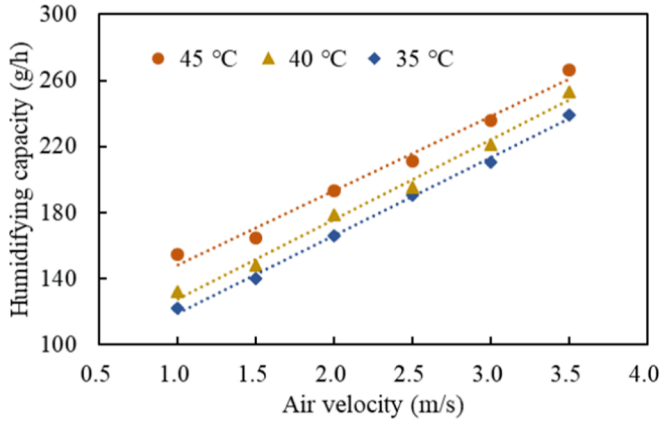


Fig. 8. Variation trend of humidification capacity with temperature and mainstream air velocity.

humidity is shown in Fig. 9. As can be seen from Fig. 9, in the case of a certain temperature and mainstream air velocity, humidification capacity increased with decreasing relative humidity. Every 5 % reduction in relative humidity, the humidification capacity increased by about 10 g/h. The humidification capacity with a relative humidity of 15 % was approximately 1.2 to 1.4 times that of a humidification capacity with a relative humidity of 30 %.

3.2. Theoretical humidification capacity calculation

The diffusion coefficient D of water molecules in air is calculated according to Eq. (2).

$$D = D_0 \left(\frac{T_m}{T_0} \right)^{\frac{3}{2}} \quad (2)$$

where, D is the diffusion coefficient of water molecules in air, D_0 is the diffusion coefficient of water molecules in air at the temperature $T_0 = 273 \text{ K}$, $D_0 = 2.2 \times 10^{-5} \text{ m}^2/\text{s}$, and T_m is the average temperature of the air in the concentration boundary layer, as determined by Eq. (3).

$$T_m = \frac{T_a + T_p}{2} \quad (3)$$

where, T_a is the air temperature before the humidification element, T_p is the water temperature on the porous plate.

The water temperature on the surface of the porous plate was not easy to measure due to the thinness of the porous plate and the small plate spacing. Thermocouples were used to monitor the water temperature inside the porous plate (shown in Fig. 5). At the same time, the wet

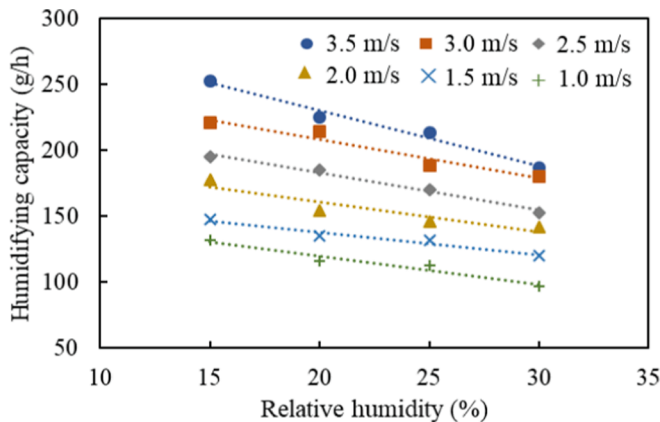


Fig. 9. Variation trend of humidification capacity with relative humidity.

bulb temperature of the air passing through the porous plate can be obtained based on the dry bulb temperature and relative humidity of the air. A comparison of the wet bulb temperature of the air with the average water temperature at the three monitoring points inside the porous plate is shown in Fig. 10. From Fig. 10, it can be seen that the fitting curve was almost a straight line passing through the origin, $R^2 = 0.9788$, which meant that the water temperature inside the porous plate was almost consistent with the wet bulb temperature of the air. Considering the fluctuation or non-uniformity of water temperature, there may be a slight difference between the water temperature inside the plate and the water temperature on the plate surface. In the calculation, a relatively stable air parameter (wet bulb temperature) was selected as the water temperature on the porous plate.

The air flows along the plane of the porous plates, and the Reynolds number Re is calculated according to Eq. (4).

$$Re = \frac{ul}{\nu} \quad (4)$$

where, Re is the Reynolds number, u is the mainstream air velocity, l is the length of the porous plate, and ν is the air viscosity, which can be obtained according to the average temperature T_m in the boundary layer.

The Schmidt number Sc is calculated by Eq. (5).

$$Sc = \frac{\nu}{D} \quad (5)$$

Since the Reynolds number Re of all experimental conditions is less than 10^5 , the Sherwood number Sh is calculated according to Eq. (6).

$$Sh = 0.664 Re^{\frac{1}{2}} Sc^{\frac{1}{3}} \quad (6)$$

The average convective mass transfer coefficient h_m can be calculated according to Eq. (7).

$$h_m = \frac{Sh \cdot D}{l} \quad (7)$$

The mass diffusion flux m_w is calculated by Eq. (8).

$$m_w = h_m (C_p - C_a) MA = A \frac{h_m}{R_w} \left(\frac{P_{ps}}{T_p} - \frac{P_a}{T_a} \right) \quad (8)$$

where, m_w is the mass diffusion flux, C_p is the concentration of water vapor on the porous plate, C_a is the concentration of water vapor in the air, M is the molar mass of water vapor, A is the area of the porous plates, R_w is the gas constant of water vapor with a value of $461.89 \text{ J/kg} \cdot \text{K}$, P_{ps} is the partial pressure of saturated water vapor at the state of temperature T_p , and P_a is the partial pressure of water vapor at the state of temperature T_a and relative humidity φ_a .

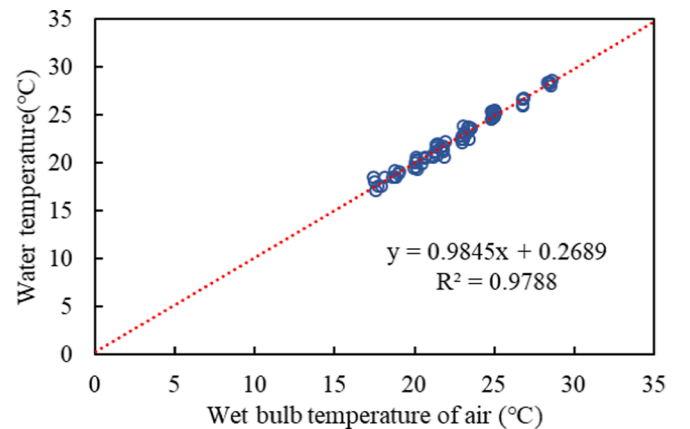


Fig. 10. Comparison between air wet bulb temperature and water temperature inside the porous plate.

The theoretical humidification capacity m_w , calculated according to Eqs. (2) to (8) based on the monitored mainstream air velocity of the experimental air duct, is compared with the measured humidification capacity H , calculated according to Eq. (1), as shown in Fig. 11. As can be seen from Fig. 11, the trends of the theoretical and measured humidification capacity were consistent, both of which increased with the increased in air temperature, the increased in mainstream air velocity, and the decreased in relative humidity. The experimental value of humidification capacity was about 1.1 to 1.2 times higher than the theoretical value. The maximum difference between the theoretical and experimental values was 51.62 g/h at a temperature of 35 °C, a relative humidity of 15 %, and a mainstream air velocity of 3.5 m/s. The relative errors for most of the conditions were within 20 %, a small number of conditions were within 30 % when the air temperature was 35 °C (Fig. 11 (a)). At an air temperature of 40 °C, the relative errors were within 15 % for most conditions and 20 % for a few conditions (Fig. 11 (b)). At an air temperature of 45 °C, the relative errors were within 10 % for most conditions and 15 % for a few conditions (Fig. 11(c)). The

theoretical values calculated by the criterion correlation can reflect the humidification capacity of the system to a certain extent.

3.3. Heat transfer quantity calculation

The Nusselt number Nu can be calculated from Eq. (9).

$$Nu = 0.664Re^{\frac{1}{2}}Pr^{\frac{1}{3}} \quad (9)$$

Where, Pr is the Prandtl number, which can be obtained from the average temperature T_m of the boundary layer.

The average convective heat transfer coefficient h can be calculated according to Eq. (10).

$$h = \frac{Nu \cdot \lambda}{l} \quad (10)$$

where, λ is the thermal conductivity, which can be obtained from the average temperature T_m of the boundary layer. Nu is the Nusselt number,

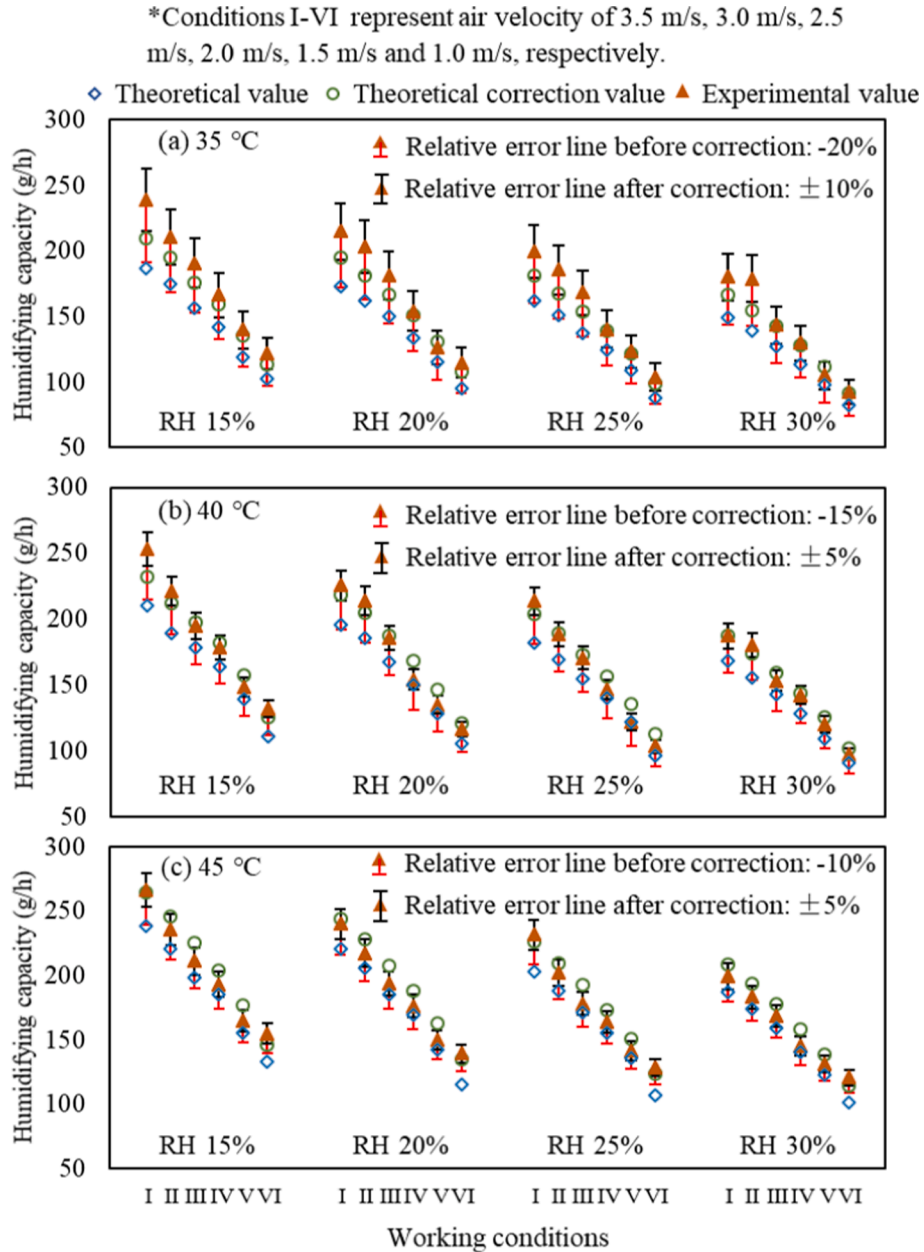


Fig. 11. Comparison between theoretical values of the criterion correlation and measured values.

l is the length of the porous plate.

The convective heat transfer quantity Q_c is calculated according to Eq. (11).

$$Q_c = hA\Delta t_m = hA \frac{(T_a - T_p) - (T_e - T_p)}{\ln \frac{T_a - T_p}{T_e - T_p}} \quad (11)$$

where, h is the average convective heat transfer coefficient, A is the area of the porous plates, Δt_m is the log-mean heat transfer temperature difference, T_a is the air temperature before the humidifying element, T_e is the air temperature after the humidification element, T_p is the water temperature on the porous plate.

The heat quantity absorption by evaporation of water, Q_e , can be calculated according to Eq. (12) based on the latent heat of water vaporization.

$$Q_e = rH \quad (12)$$

where, r is the latent heat of water vaporization, which can be obtained

from the water temperature on the porous plate (the wet bulb temperature of the air in this paper). H is the humidification capacity calculated by weighting method.

Theoretically, the heat quantity required for water evaporation is provided by the convective heat transfer quantity of air. Fig. 12 is the comparison result of the convective heat transfer quantity of air Q_c (theoretical value calculated on the basis of the monitored mainstream air velocity in the experimental air duct) and the heat quantity absorption by evaporation of water Q_e (experimental value). As can be seen from Fig. 12, the convective heat transfer quantity and the heat quantity absorption by evaporation showed the same change trend, both of which increased with the increased in air temperature, decreased in relative humidity, and increased in mainstream air velocity. The difference between the convective heat transfer quantity and the heat quantity absorption by evaporation was not significant. The relative errors range for most conditions were within 20 %, and the heat quantity absorption by evaporation was greater than the convection heat transfer quantity, which was about 1.1 to 1.2 times higher.

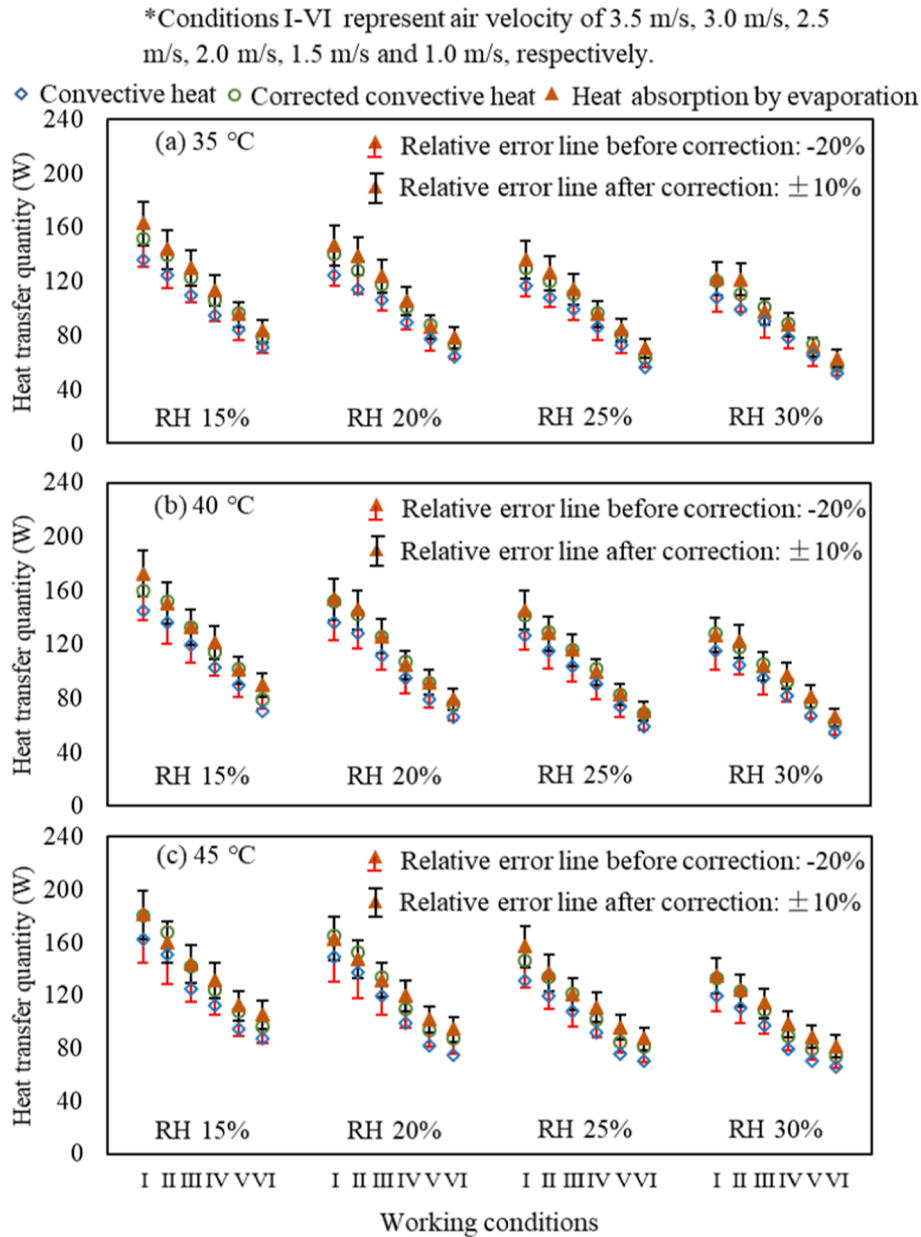


Fig. 12. Comparison of convective heat transfer and evaporative heat absorption.

3.4. Simulation analysis

Due to the fact that the theoretical humidification capacity (criterion correlation) and theoretical heat transfer quantity (convective heat transfer) were calculated based on the mainstream air velocity of the experimental air duct, the air velocity between porous plates was actually different from the mainstream air velocity. The velocity u in Eq. (4) should be the mainstream air velocity within the porous plate gap rather than the experimental air duct. Due to the small spacing of the porous plates, the air velocity measurement device cannot accurately monitor the air velocity within the plate gap. In order to investigate the reasons for the difference between experimental and theoretical values, a simulation study was conducted on the flow field inside the experimental duct. The flow field simulation diagram is shown in Fig. 13. The inlet boundary condition was selected as the velocity inlet, the outlet boundary condition was the pressure outlet, and all other walls were non-slip walls. The number of grids used for simulation after grid independence testing was 8 million. The flow equation was chosen as the k - ϵ turbulence equation, and the algorithm was chosen as the coupled algorithm for simulation calculation.

The results of the flow field simulation are shown in Fig. 14. As can be seen from Fig. 14, when the air flowed through the porous plates, the color at the cross-section of the porous plates was significantly darker than that at the cross section in front of the porous plates, indicating that the air velocity in the gap between the porous plates was greater than that in front of the porous plates (the mainstream air velocity of the experimental air duct), which might led to the experimental values of humidification capacity and heat transfer quantity being greater than the theoretical values.

The average air velocity in the cross-section (characteristic cross-section, Fig. 13) between two porous plates was taken as the mainstream air velocity in the porous plate gap. Table 2 shows the simulated air velocity values.

Comparisons of the theoretical corrected humidification capacity, corrected convective heat transfer quantity obtained with the simulated velocities in Table 2 with the measured values are also shown in Figs. 11 and 12. As can be seen from Fig. 11, at an air temperature of 35 °C, most of the relative errors in humidification capacity had been reduced to within 10 % (Fig. 11(a)). When the air temperature was 40 °C and 45 °C, most of the relative errors of humidification capacity were within 5 %, and a few relative errors were within 10 % (Fig. 11(b) and (c)). As can be seen in Fig. 12, the relative errors in heat transfer quantity were within 10 % for most of the conditions and within 15 % for very few conditions.

The corrected theoretical values obtained with the simulated mainstream air velocities were more accurate. In addition to certain unknown factors, the reasons for errors may also include instrument accuracy and measurement errors. At the same time, the humidification capacity is also related to the humidification material's own pore parameters (such

as porosity, pore size distribution, contact angle, etc.), and moisture absorption and release characteristics. The theoretical calculation in this article only considers the criterion correlation method in the mass transfer process of flat plates, without considering the influence of these specific factors.

3.5. Comparison with previous studies

The humidification performance of the porous fiber materials used in this paper was compared with that of similar materials in previous studies, as shown in Table 3. Previous studies mostly focused on the folding shaped humidification element shown in reference [13] or the packing shaped humidification element shown in reference [41], and few of the parallel spaced arrangement humidification elements in currently studied. A comparison of the current study with previous studies in terms of materials used, material sizes and quantities, humidification element sizes, inlet airflow, inlet temperature and humidity, and humidification capacity is presented in Table 3. As can be seen in Table 3, the micro-/nanofiber composite membrane used in reference [13] had a higher humidification capacity, not only due to its large humidification area, but also attributed to the fact that it had a nano-scale fiber diameter, which conferred a superior humidification performance. The date palm tree fibers used in reference [41] was a natural fiber, and although it had a large humidification area and high inlet air volume, its humidification performance was lower.

Different material properties lead to different moisture transfer properties and heat transfer rates, and in-depth theoretical and experimental comparative studies are needed to explore the reasons [25]. The humidification materials in the current study are different in thickness, permeability, and moisture absorption capacity than those in previous studies, and may not be directly comparable to those in references [13] and [41]. However, porous plate humidification materials have played a positive role in addressing the practical application of low indoor relative humidity. The author has applied it to the fan coil and tested it [18]. Compared with the non-use of humidifying elements, the use of humidifying elements increases the indoor relative humidity by 34.1 %, while the temperature is only reduced by 0.3 °C, which can significantly increase the indoor humidity on the premise of meeting the indoor heat production.

4. Conclusion

In order to solve the problem of low indoor relative humidity in winter in severe cold regions and promote the application of porous plate humidification elements, a performance test bench was built to analyze the humidification performance of porous plates, and the conclusions were as follows.

- (1) The wicking height of the absorbent fiberboard used in this paper reached 72 mm within 60 s. Under certain working conditions, the evaporation process of water was continuous and stable. The humidification effect of porous plates was excellent. At a relative humidity of 15 %, mainstream air velocity of 3.5 m/s, and air temperature of 45 °C, the maximum humidification capacity of the system reached 266.4 g/h.
- (2) The humidification capacity was related to the air temperature, relative humidity and mainstream air velocity. The higher the air temperature, the lower the relative humidity, and the greater the mainstream air velocity, the greater the humidification capacity. For every 5 °C increased in temperature, the humidification capacity increased by about 10 g/h. For every 5 % decreased in relative humidity, the humidification capacity increased by approximately 10 g/h. The mainstream air velocity increased by 0.5 m/s, and the humidification capacity increased by about 20 g/h.

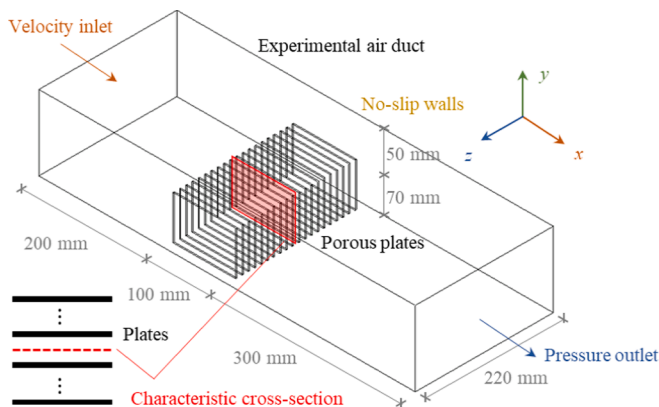


Fig. 13. Schematic diagram of flow field simulation.

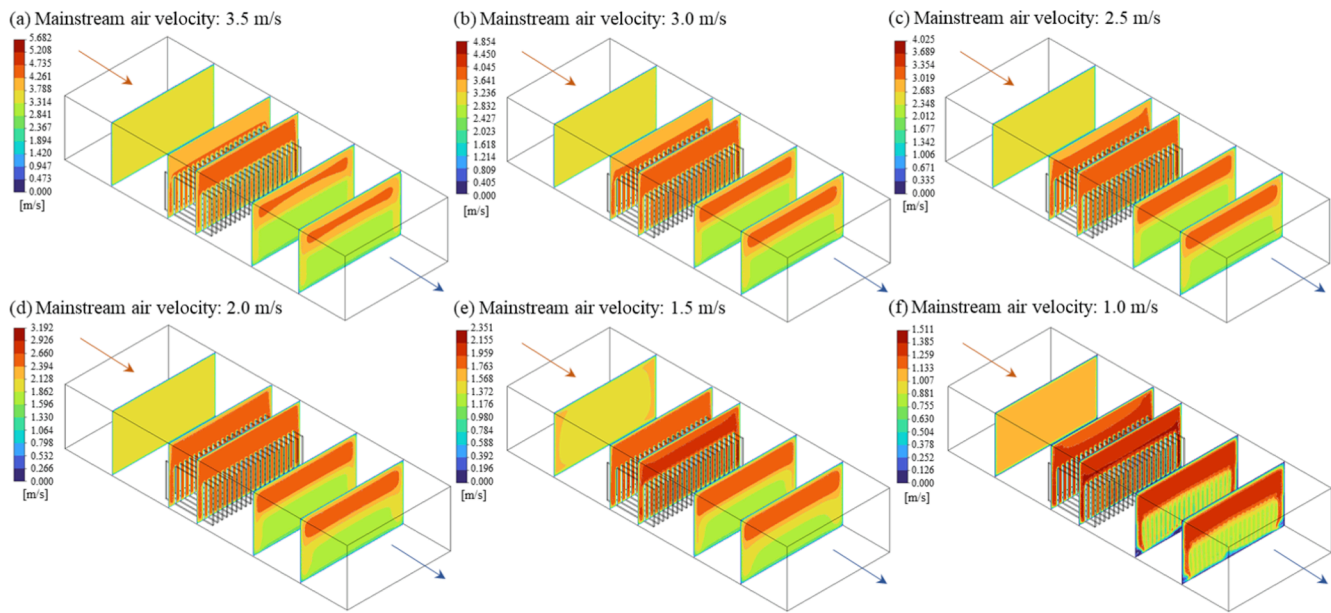


Fig. 14. Flow field simulation results of experimental ducts.

Table 2
Simulated values of mainstream air velocity in the gap of the porous plate.

Air velocity of the experimental duct (m/s)	3.5	3.0	2.5	2.0	1.5	1.0
Air velocity of characteristic cross-section (m/s)	4.367	3.767	3.164	2.557	1.945	1.316

(3) The variation trend of humidification capacity and heat transfer quantity calculated by the criterion correlation was consistent with the measured humidification capacity and heat transfer quantity, and the measured values were all greater than the theoretical values. The relative errors of the humidification capacity for most conditions were within 20 % (the air temperature was 35 °C), 15 % (the air temperature was 40 °C), 10 % (the air

temperature was 45 °C). The relative errors in heat transfer quantity under most working conditions was within 20 %.

(4) The CFD simulation results showed that the air velocity in the gap between the porous plates was larger than the mainstream air velocity of the experimental air duct in the theoretical calculation might be one of the reasons for the errors. The relative errors of the theoretical humidification capacity and heat transfer quantity calculated by the criterion correlation according to the simulated air velocity and the measured humidification capacity and heat transfer quantity were reduced. The relative errors of humidification capacity were reduced to 10 % (35 °C) and 5 % (40 °C, 45 °C) for most conditions, and the relative errors of heat transfer quantity were within 10 % for most conditions.

The porous fiber material studied in this paper has a good wicking absorption capacity, and the top spray of the traditional direct evaporation cooling system can be eliminated in the structure of the humidification system, and only the water pan is set below. And the fiber

Table 3
Comparison between current study and previous studies.

	Current study	Chen et al. [13]	Sellam et al. [41]
Materials material size	Polyester fiber Height: 100 mm Depth: 100 mm Thickness: 2 mm	Micro-/nanofibers Height: 130 mm Depth: 30 mm Thickness: 580 ± 10 μm	Date palm tree fibers —
Element size (mm)	192 × 100 × 100	227 × 30 × 130	300 × 30 × 300
Material quantity	Plate spacing: 8 mm 20 pieces	Fold spacing: 5 mm About 40 pieces*	311.5 g
Shape schematic*			
Air volume	3.5 m/s (323 m³/h, 0.110 kg/s)	330 m³/h	0.165 kg/s
Air temperature	45 ± 0.5 °C	25 ± 2 °C	31 ~ 42 °C*
Air relative humidity	15 ± 2 %	30 ± 5 %	15 ~ 50 %*
Humidification capacity	266 g/h	514 g/h	Average: 220 g/h*

* Indicates that it was not explicitly stated in the reference and was obtained by organizing the literature according to its content.

material is hard, no moving parts, suitable for miniaturized production, can be placed in the fan coil rather than the air handling device. The porous plate can satisfy the original heating capacity while providing humidification capacity, which provides a feasible idea for solving the dual needs of indoor heating and humidification in winter to the users without centralized air conditioning.

CRedit authorship contribution statement

Jiayu Ye: Conceptualization, Data curation, Investigation, Writing – original draft, Writing – review & editing. **Hui Yan:** Investigation, Writing – review & editing. **Long Ni:** Conceptualization, Funding acquisition, Methodology, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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References

- [1] Z. Li, Z. Di, M. Chang, et al., Study on the influencing factors on indoor PM_{2.5} of office buildings in Beijing based on statistical and machine learning methods, *Journal of Building Engineering* 66 (2023) 105240.
- [2] D. Amaripadath, R. Rahif, M. Velickovic, et al., A systematic review on role of humidity as an indoor thermal comfort parameter in humid climates, *J. Build. Eng.* (2023) 106039.
- [3] T. Psomas, D. Teli, S. Langer, et al., Indoor humidity of dwellings and association with building characteristics, behaviors and health in a northern climate, *Build. Environ.* 198 (2021) 107885.
- [4] P. Mecenas, R.T.R.M. Bastos, A.C.R. Vallinoto, et al., Effects of temperature and humidity on the spread of COVID-19: A systematic review, *PLoS One* 15 (9) (2020) e0238339.
- [5] R. Pal, S. Sarkar, A. Mukhopadhyay, Influence of ambient conditions on evaporation and transport of respiratory droplets in indoor environment, *Int. Commun. Heat Mass Transfer* 129 (2021) 105750.
- [6] M. Moriyama, W.J. Hugentobler, A. Iwasaki, Seasonality of respiratory viral infections, *Annual Review of Virology* 7 (2020) 83–101.
- [7] Kudo E, Song E, Yockey L J, et al. Low ambient humidity impairs barrier function and innate resistance against influenza infection[J]. *Proceedings of the National Academy of Sciences*, 2019, 116(22): 10905-10910.
- [8] P. Wolkoff, Dry eye symptoms in offices and deteriorated work performance—A perspective, *Build. Environ.* 172 (2020) 106704.
- [9] P. Wolkoff, “Healthy” eye in office-like environments, *Environ. Int.* 34 (8) (2008) 1204–1214.
- [10] P. Wolkoff, K. Azuma, P. Carrer, Health, work performance, and risk of infection in office-like environments: The role of indoor temperature, air humidity, and ventilation, *International Journal of Hygiene and Environmental Health* 233 (2021) 113709.
- [11] P. Wolkoff, Indoor air humidity, air quality, and health—An overview, *Int. J. Hyg. Environ. Health* 221 (3) (2018) 376–390.
- [12] J. Yao, J. Li, Y. Zhao, et al., Characterization of granular electrostatics generation, *Powder Technol.* 363 (2020) 74–85.
- [13] J. Chen, J. Mai, C. Wang, et al., Biomimetic aligned micro-/nanofibrous composite membranes with ultrafast water transport and evaporation for efficient indoor humidification, *ACS Appl. Mater. Interfaces* 14 (1) (2021) 1983–1993.
- [14] Z. Yang, L.A. Chen, C. Yang, et al., Portable ultrasonic humidifier exacerbates indoor bioaerosol risks by raising bacterial concentrations and fueling pathogenic genera, *Indoor Air* 32 (1) (2022) e12964.
- [15] K. Nordström, D. Norbäck, R. Axelsson, Effect of air humidification on the sick building syndrome and perceived indoor air quality in hospitals: a four month longitudinal study, *Occup. Environ. Med.* 51 (10) (1994) 683–688.
- [16] S.S. Salins, S.V.K. Reddy, S. Kumar, et al., Mathematical model-based performance assessment of a centrifugal humidifier under varying operating conditions, *Int. Commun. Heat Mass Transfer* 144 (2023) 106786.
- [17] R. Tu, J. Li, Y. Hwang, Performance analysis of desiccant wheels assisted fresh air humidifiers in winter using natural gas boilers: Applied in cold and dry climate regions, *Int. J. Refrig* 119 (2020) 24–36.
- [18] J. Ye, X. Liu, L. Xu, et al., Experimental study on the heating and humidifying performance of fan coil units with humidification modules in severe cold regions, *Energ. Buildings* 276 (2022) 112500.
- [19] Z. Feng, X. Zhou, S. Xu, et al., Impacts of humidification process on indoor thermal comfort and air quality using portable ultrasonic humidifier, *Build. Environ.* 133 (2018) 62–72.
- [20] M. Duan, Y. Lu, Y. Li, et al., Indoor dryness and humidification-induced arsenic inhalation exposure above 4200 m in Ngari, China *Indoor Air* 32 (10) (2022) e13133.
- [21] Y. Lu, Practical Design Manual for Heating and Air Conditioning, 2nd Edition, China Architecture & Building Press, Beijing, 2008 in Chinese.
- [22] F. Yang, X.N. Wu, J.J. Peng, et al., Investigation on humidification performance of silica gel rotary wheel system in winter, *Build. Environ.* 183 (2020) 107064.
- [23] T. Gunhan, V. Demir, A.K. Yagcioglu, Evaluation of the suitability of some local materials as cooling pads, *Biosyst. Eng.* 96 (3) (2007) 369–377.
- [24] Y. Yang, G. Cui, C.Q. Lan, Developments in evaporative cooling and enhanced evaporative cooling—A review, *Renew. Sustain. Energy Rev.* 113 (2019) 109230.
- [25] Y. Chen, X. Huang, T. Sun, et al., Experimental study of plant fiber-polymer composite for indirect evaporative cooler application, *Appl. Therm. Eng.* 199 (2021) 117543.
- [26] P. Xu, X. Ma, X. Zhao, et al., Experimental investigation on performance of fabrics for indirect evaporative cooling applications, *Build. Environ.* 110 (2016) 104–114.
- [27] N.H. Kim, Performance of an indirect evaporative cooler (IEC) made of PET/cellulose composite sheet as wetting media, *Appl. Therm. Eng.* 186 (2021) 116492.
- [28] C.I. Su, J.X. Fang, X.H. Chen, et al., Moisture absorption and release of profiled polyester and cotton composite knitted fabrics, *Text. Res. J.* 77 (10) (2007) 764–769.
- [29] J.A. Kicińska, E. Bogacz, M. Zimniewska, Review of natural fibers. part I- vegetable fibers, *J. Nat. Fibers* 9 (3) (2012) 150–167.
- [30] P.A. Doğramacı, D. Aydın, Comparative experimental investigation of novel organic materials for direct evaporative cooling applications in hot-dry climate, *Journal of Building Engineering* 30 (2020) 101240.
- [31] Y. Han, Z. Jiang, Z. Liang, et al., Full pore size fractal characteristics of longmaxi formation shale in luzhou-changing area and its influence on seepage, *Energy Fuel* 37 (15) (2023) 11083–11096.
- [32] W. Zheng, Y. Li, Y. Zhu, et al., Preparation of a novel structure-designed porous ceramsite for water retention, *ACS Appl. Mater. Interfaces* 15 (36) (2023) 42753–42763.
- [33] L.M. Beleuk à Moungam, P.N. Lemougna, R.C. Kaze, et al., Synthesis of volcanic ash-based porous inorganic polymers using biomass as pore inducing agent: phase evolution and descriptive microstructure, *SILICON* (2022): 1–14.
- [34] A. Bada, H. Khelafi, A.M. Mokhtari, et al., Experimental study of the hygrothermal behavior of an earth/porous pipe/air heat exchanger (EPPAHE) for building cooling needs in the context of the hot and dry Saharan climate, *Thermal Science and Engineering Progress* 37 (2023) 101598.
- [35] H. Ulusu, H.Y. Aruntaş, A.B. Gültekin, et al., Mechanical, durability and microstructural characteristics of Portland pozzolan cement (PPC) produced with high volume pumice: Green, cleaner and sustainable cement development, *Constr. Build. Mater.* 378 (2023) 131070.
- [36] J. Lv, H. Xu, T. Xu, et al., Study on the performance of a unit dew-point evaporative cooler with fibrous membrane and its application in typical regions, *Case Studies in Thermal Engineering* 24 (2021) 100881.
- [37] D. Bishoyi, K. Sudhakar, Experimental performance of a direct evaporative cooler in composite climate of India, *Energ. Buildings* 153 (2017) 190–200.
- [38] S.A. Nada, H.F. Elattar, M.A. Mahmoud, et al., Performance enhancement and heat and mass transfer characteristics of direct evaporative building free cooling using corrugated cellulose papers, *Energy* 211 (2020) 118678.
- [39] J. Zhang, S. Hong, B. Dong, et al., Water distribution modelling of capillary absorption in cementitious materials, *Construction and Building Materials* 216 (2019) 468–475.
- [40] Chu J, Huang X, Sun T, et al. Hydrophilic study of dew point indirect evaporative cooler wet channel side material[J]. *Cotton Textile Technology*, 2018, 46(1). (in Chinese).
- [41] S.H. Sellam, A. Moummi, C.E. Mehdid, et al., Experimental performance evaluation of date palm fibers for a direct evaporative cooler operating in hot and arid climate, *Case Studies in Thermal Engineering* 35 (2022) 102119.